

WEEKLY PROJECT REPORT

on

FPGA Implementation of a 100-Neuron Neural Network Using Verilog HDL

Submitted as part of the Internship Programme

Submitted By

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June 2026

Declaration

I hereby declare that the project report entitled “**FPGA Implementation of a 100-Neuron Neural Network Using Verilog HDL**” is a record of the work carried out by me during my internship under the guidance of **Dr. Surya Teja Srinivasa Padala**, Department of Electronics and Communication Engineering, National Institute of Technology Tiruchirappalli.

Place: Tiruchirappalli

Date: _____

Abstract

Multi-neuron neural networks are computational models composed of multiple interconnected neurons that work collectively to process input information, extract meaningful patterns, and generate output responses. This project presents the design and implementation of a hardware-based neural network consisting of three input neurons, a hidden layer containing 100 neurons, and two output neurons using Verilog Hardware Description Language (HDL) on the Basys 3 FPGA platform.

The proposed architecture employs weighted-sum computations and threshold-based spike generation within the hidden layer. Each hidden neuron processes the input signals using unique weight and bias values, producing spike responses based on neuron activation. The collective activity of the hidden layer is represented through a population spike count mechanism. The generated spike count is accumulated and displayed on the seven-segment display available on the FPGA board, enabling real-time observation of network activity.

The design was developed, synthesized, and implemented using Xilinx Vivado Design Suite. Functional verification was carried out through simulation, followed by hardware validation on the Basys 3 FPGA board. Resource utilization and power analysis were also performed to evaluate the efficiency of the implementation. Experimental results demonstrate the successful operation of the neural network and validate the feasibility of implementing neural computation using FPGA-based hardware architectures.

Keywords: Artificial Neural Network, FPGA, Verilog HDL, Basys 3, Hidden Layer, Spike Generation, Neural Computation, Vivado.

Chapter 1

Introduction

Neural networks are computational models inspired by the biological nervous system. They consist of interconnected neurons that process input information and generate meaningful outputs. By combining the responses of multiple neurons, neural networks can identify patterns and perform decision-making tasks.

With the growing demand for real-time processing and hardware acceleration, Field Programmable Gate Arrays (FPGAs) offer advantages such as parallel computation, reconfigurability, low latency, and efficient resource utilization.

This project focuses on the implementation of a multi-neuron neural network on the Basys 3 FPGA board using Verilog Hardware Description Language (HDL). The proposed architecture consists of three input neurons, a hidden layer containing one hundred neurons, and two output neurons. Each hidden neuron performs weighted-sum computations using predefined weights and bias values. The computed activation is compared against a threshold to generate spike responses, mimicking the behavior of biological neurons.

The spike outputs generated by the hidden layer are accumulated to determine the overall network activity. This activity is represented through a population spike count, which is displayed on the seven-segment display of the Basys 3 FPGA board. The implementation demonstrates how neural computation can be efficiently realized using digital hardware.

The primary objective of this work is to design, simulate, synthesize, and implement a 100-neuron neural network architecture on FPGA hardware and evaluate its functionality through simulation results, hardware validation, resource utilization analysis, and power consumption analysis.

1.1 Objectives

- To design a neural network architecture consisting of three inputs, one hidden layer, and two outputs.
- To implement a hidden layer containing 100 neurons using Verilog HDL.
- To perform weighted-sum computation and threshold-based spike generation.
- To calculate and display the population spike count on the seven-segment display.
- To validate the design through simulation and FPGA implementation.
- To analyze the resource utilization and power consumption of the design.

Chapter 2

Methodology

2.1 Proposed Architecture

The proposed neural network architecture consists of three input neurons, a hidden layer containing 100 neurons, and two output neurons. The design is implemented using Verilog HDL and deployed on the Basys 3 FPGA platform. The hidden layer performs the primary computation of the network by processing the input signals using predefined weights and bias values.

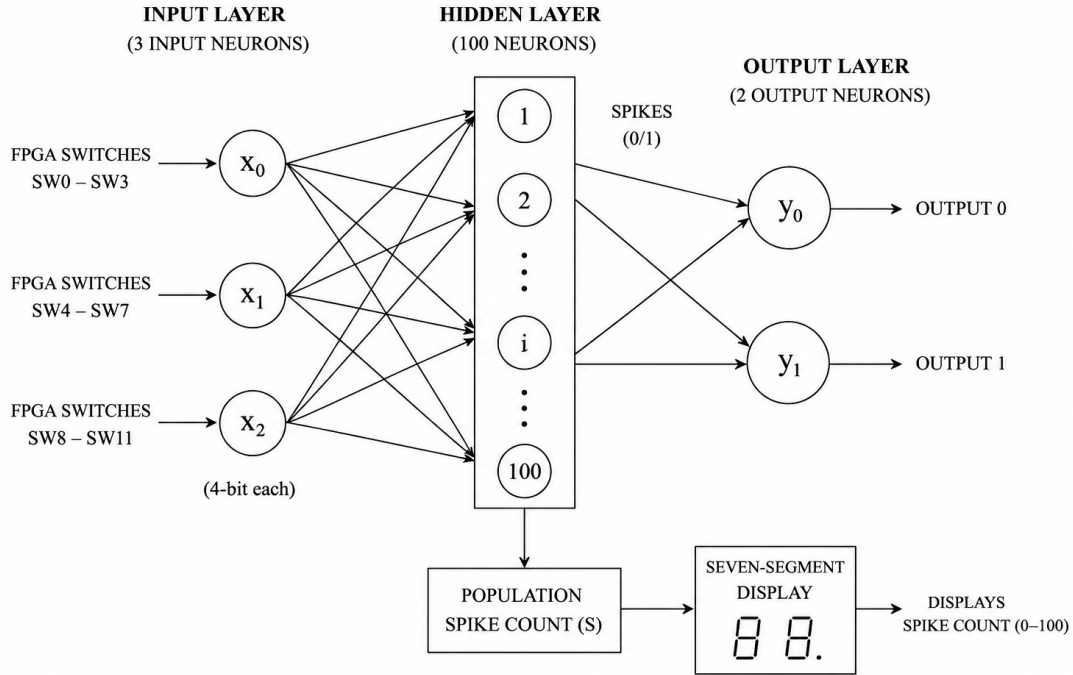


Figure 2.1: Proposed 100-Neuron Neural Network Architecture

2.2 Input Layer

The input layer consists of three input neurons represented by x_0 , x_1 , and x_2 . These inputs are provided through the switches available on the Basys 3 FPGA board. Each input is represented

using a 4-bit binary value and is connected to all neurons in the hidden layer.

2.3 Hidden Layer

The hidden layer contains 100 neurons. Each neuron receives the same set of inputs but processes them using different weight and bias values. This allows different neurons to respond differently to the same input condition.

The activation value of the i^{th} hidden neuron is calculated using a weighted-sum operation as follows:

$$s_i = x_0w_{i0} + x_1w_{i1} + x_2w_{i2} + b_i \quad (2.1)$$

where,

- x_0 : Input Neuron 0
- x_1 : Input Neuron 1
- x_2 : Input Neuron 2
- w_{i0} : Weight connecting x_0 to hidden neuron i
- w_{i1} : Weight connecting x_1 to hidden neuron i
- w_{i2} : Weight connecting x_2 to hidden neuron i
- b_i : Bias of hidden neuron i
- s_i : Activation value of hidden neuron i

2.4 Spike Generation

After computing the activation value, each hidden neuron compares the result with a predefined threshold value. In this implementation, the threshold is set to 50.

$$T = 50 \quad (2.2)$$

A spike is generated whenever the activation value exceeds the threshold.

$$spike_i = \begin{cases} 1, & s_i > T \\ 0, & s_i \leq T \end{cases} \quad (2.3)$$

A spike value of 1 indicates that the neuron is active, while a value of 0 indicates that the neuron remains inactive.

2.5 Population Spike Count

The spikes generated by all hidden neurons are accumulated to determine the overall activity of the hidden layer. This process is referred to as population spike counting.

$$S = \sum_{i=1}^{100} spike_i \quad (2.4)$$

where,

- $speak_i$: Spike generated by hidden neuron i
- S : Total population spike count

The value of S can range from 0 to 100, depending on the number of neurons that generate spikes for a given input condition.

2.6 Output Layer

The output layer consists of two output neurons. These outputs are derived from the overall hidden-layer activity represented by the population spike count. The outputs indicate the final response of the neural network based on the processed input information.

2.7 Display Unit

The total spike count is displayed on the seven-segment display available on the Basys 3 FPGA board. A seven-segment display driver converts the binary spike count into decimal digits and continuously refreshes the display, enabling real-time visualization of hidden-layer activity.

2.8 Hardware Implementation Flow

The complete design flow consists of the following steps:

1. Input acquisition through FPGA switches.
2. Weighted-sum computation in 100 hidden neurons.
3. Threshold comparison and spike generation.
4. Population spike accumulation.
5. Output neuron evaluation.
6. Display of spike count on the seven-segment display.

Chapter 3

Results and Discussion

3.1 RTL Schematic

The proposed 100-neuron neural network was synthesized using Xilinx Vivado Design Suite. The generated RTL schematic confirms the successful implementation of the complete neural network architecture, including the input interface, hidden-layer neurons, spike generation logic, spike accumulation block, output generation circuitry, and seven-segment display driver.

The RTL view verifies that the design has been correctly translated into FPGA hardware resources. The schematic also demonstrates the modular structure of the implementation, enabling scalability for larger neural network architectures in future work.

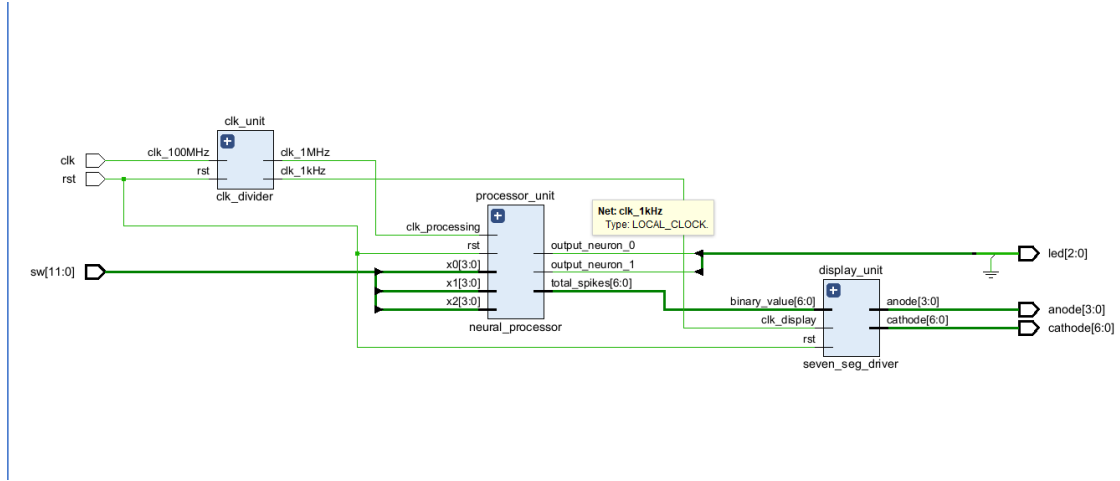


Figure 3.1: RTL Schematic of the Proposed Neural Network

3.2 Simulation Results

Functional simulation was performed to verify the correctness of the proposed design before FPGA implementation. Various combinations of input values were applied, and the corresponding outputs were observed using simulation waveforms.

The simulation results confirmed that the hidden neurons correctly perform weighted-sum computations using the assigned weights and bias values. The computed activation values are compared against a predefined threshold, and spikes are generated whenever the threshold condition is satis-

fied.

The spike outputs generated by all hidden neurons are accumulated to obtain the total population spike count. The waveform results verified the correct operation of the hidden layer, spike generation logic, spike counter, and output generation blocks.

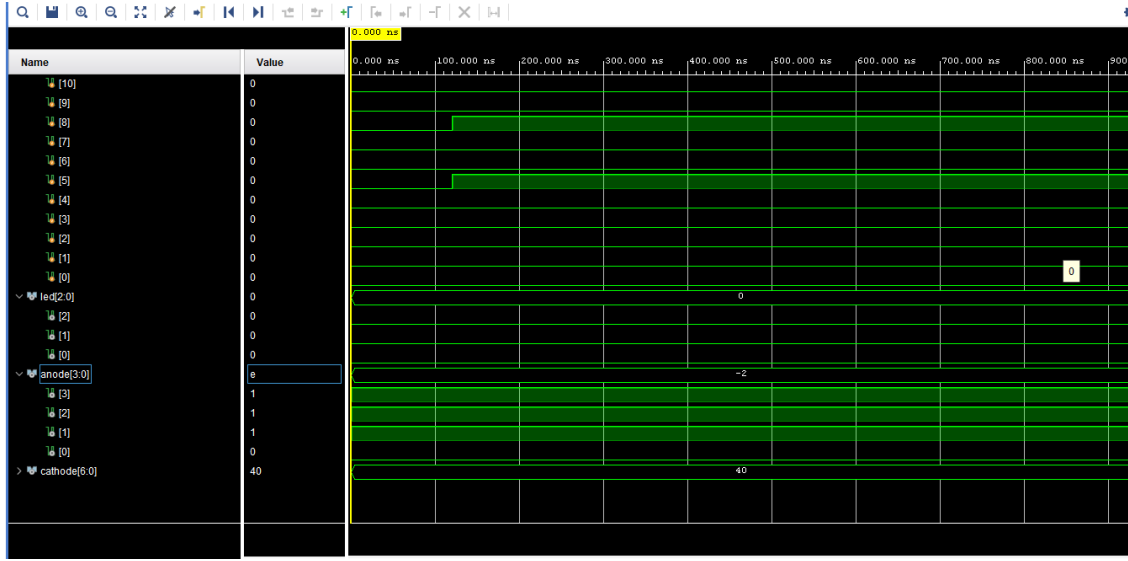


Figure 3.2: Simulation Waveform of the Neural Network

3.3 FPGA Implementation Results

After successful simulation and synthesis, the design was implemented on the Basys 3 FPGA board. Input values were provided through the onboard switches, and the resulting spike count was displayed on the seven-segment display.

For the input condition:

$$x_0 = x_1 = x_2 = 4$$

the FPGA implementation produced a population spike count of:

$$S = 43$$

The displayed value indicates that 43 hidden neurons generated spikes for the given input combination. Since the hidden layer contains 100 neurons, the result shows that 43 neurons became active and crossed the threshold condition.

The hardware output matched the expected simulation behavior, confirming the correctness of the FPGA implementation.

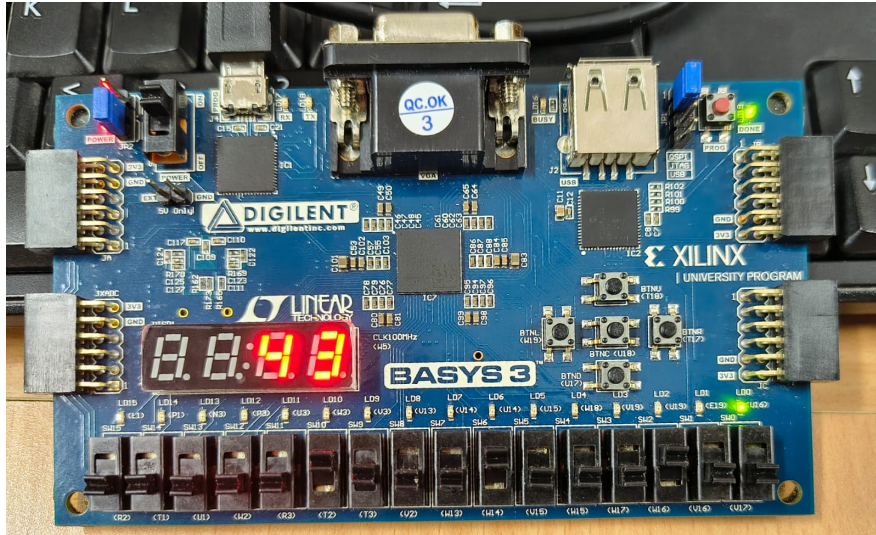


Figure 3.3: FPGA Implementation Result Showing a Spike Count of 43

1. Power Report (0.094 W)

- **Ultra-Low Consumption:** The system draws only **94 mW** total power, which is less than a tenth of a watt.
- **I/O Dominance:** About 88% of dynamic power is used for driving external pins (7-segment display and LEDs), while internal neural computation consumes minimal power.
- **Thermal Performance:** The chip operates at approximately **25.5°C**, indicating negligible heating under normal conditions.

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: 0.094 W
Design Power Budget: Not Specified
Process: typical
Power Budget Margin: N/A
Junction Temperature: 25.5°C
Thermal Margin: 74.5°C (14.8 W)
Ambient Temperature: 25.0 °C
Effective θ_{JA} : 5.0°C/W
Power supplied to off-chip devices: 0 W
Confidence level: Low
[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power

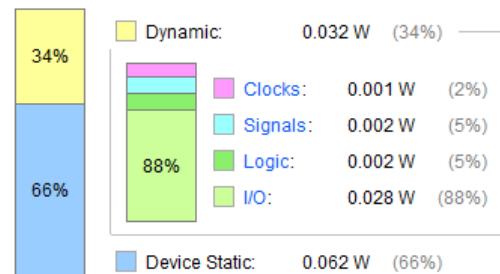
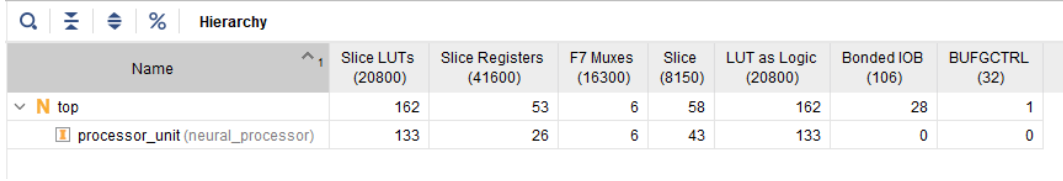


Figure 3.4: Power Analysis Report

2. Area Utilization Report (<1% Used)

- **Tiny Resource Footprint:** The design uses only **162 LUTs** and **53 registers**.
- **Time-Multiplexing Efficiency:** A single neuron processing unit is reused 100 times instead of instantiating 100 hardware neurons, resulting in extremely low FPGA resource usage (<1% of device capacity).

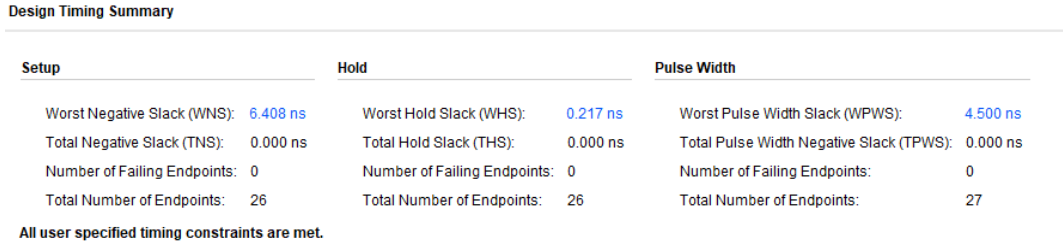


Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	Slice (8150)	LUT as Logic (20800)	Bonded IOB (106)	BUFGCTRL (32)
top	162	53	6	58	162	28	1
processor_unit (neural_processor)	133	26	6	43	133	0	0

Figure 3.5: Utilization Report

3. Timing & Delay Report

- **Timing Closure Achieved:** No timing violations detected (**TNS = 0.000 ns**).
- **Positive Slack:** Worst Negative Slack (WNS) is **+6.408 ns**, indicating strong timing margin.
- **Performance Insight:** The critical path delay is approximately **3.592 ns**, meaning the design operates well within the 10 ns clock period and has nearly $3\times$ timing headroom.



Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 6.408 ns	Worst Hold Slack (WHS): 0.217 ns	Worst Pulse Width Slack (WPWS): 4.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 26	Total Number of Endpoints: 26	Total Number of Endpoints: 27

All user specified timing constraints are met.

Figure 3.6: Timing Analysis Report

3.4 Discussion

The proposed 100-neuron neural network was successfully designed, simulated, synthesized, and implemented on the Basys 3 FPGA platform. The hidden layer correctly performed weighted-sum computations and threshold-based spike generation. The spike accumulation block accurately computed the total population spike count and displayed the result on the seven-segment display.

For the input combination where all three inputs were assigned a value of 4, the network produced a spike count of 43. This indicates that 43 hidden neurons became active for the applied input condition. The simulation results, FPGA implementation results, resource utilization analysis, and power analysis collectively validate the functionality and efficiency of the proposed architecture.

Overall, the project demonstrates the feasibility of implementing multi-neuron neural network architectures using Verilog HDL on FPGA hardware platforms.